

Control in biology: topics in bacterial growth

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Joint work with

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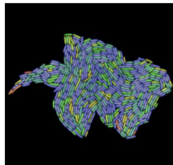
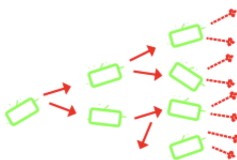
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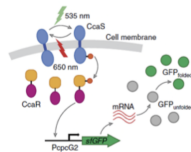
Tokyo, August 2023

Model definition

Optimal control of microbial cells (ANR Maximic, PI H. de Jong)



Stewart et al. (2005), *PLoS Biol.*, 3(2): e45



Miliadis-Angelis et al. (2019), *Nat Commun.*, 7:12546

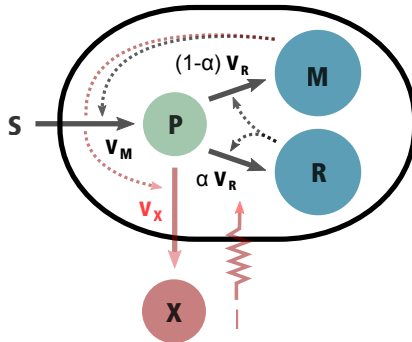
- ▶ Self-replicator model: biomass maximisation (Giordano *et al*'16)
- ▶ Genetic engineering of a strain of bacteria: light-induced control of growth
- ▶ New pathway to produce a metabolite of interest (Egorov *et al*'18)
- ▶ Competition between biomass maximisation and metabolite production

Yabo, A.; Caillaud, J.-B.; Gouzé, J.-L. Optimal bacterial resource allocation: metabolite production in continuous bioreactors. *Math. Biosci. Eng.* **17** (2020), no. 6, 7074–7100

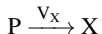
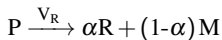
Yabo, A. G.; Caillaud, J.-B.; Gouzé, J. L.; de Jong, H.; Mairet, F. Dynamical analysis and optimization of a generalized resource allocation model of microbial growth. *SIAM J. Appl. Dyn. Syst.* **21** (2022), no. 1, 137–165

Model definition

Self-replicator model



The model represents three chemical macroreactions between substrate (S), precursors (P), genetic machinery/ribosomes (R), metabolic machinery (M) and produced metabolite (X):



New control: $u(t) := \alpha(t)I(t)$

Model definition

Continuous stirred-tank reactor dynamics

In terms of concentrations s , p , r , x (and using $m + r = 1$) in the CSTR,

$$S_1 : \begin{cases} \dot{s} = D(s_{in} - s) - v_M(s, 1 - r)\mathcal{V}, \\ \dot{p} = v_M(s, 1 - r) - v_X(p, 1 - r) - \mu(p, r)(p + 1), \\ \dot{r} = (u - r)\mu(p, r), \\ \dot{x} = v_X(p, 1 - r)\mathcal{V} - Dx, \\ \dot{\mathcal{V}} = (\mu(p, r) - D)\mathcal{V}, \end{cases}$$

where $\mathcal{V}$ is the volume of the cell population, D is the dilution rate, s_{in} is the substrate inflow, and

$$\mu(p, r) := \beta v_R(p, r)$$

(β = inverse of cytoplasmic density) is the growth rate of the bacterial population.

Assumption. The synthesis rates v_M , v_R and v_X are *monotone increasing kinetics* (e.g. Michaelis-Menten, $v_R(p, r) := k_R r p / (K_R + p)$) and

$$v_X(p, 1 - r) = c(r) v_R(p, r).$$

Asymptotic behaviour

Mass conservation

Proposition. The set

$$\Gamma = \left\{ (s, p, r, x, \mathcal{V}) \in \mathbf{R}^5 : s_{in} \geq s \geq 0, p \geq 0, x \geq 0, 1 \geq r \geq 0, \mathcal{V} \geq 0 \right\}$$

is positively invariant for the initial value problem.

We assume $\mathcal{V}(0) > 0$ and $r(0) > 0$ (no growth otherwise).

Proposition. The ω -limit set of any solution of system S_1 with constant input $u(t) = \bar{u}$ lies in the set

$$\Omega := \left\{ (s, p, r, x, \mathcal{V}) \in \mathbf{R}^5 : s + \left(p + \frac{r}{\bar{u}} \right) \mathcal{V} + x = s_{in} \right\}.$$

This result is used to define a limit system and reduce the dimension by two.

Asymptotic behaviour

Limiting system

As $r(t) \rightarrow \bar{u}$ when $t \rightarrow \infty$ and since x can be eliminated (Ω), we define

$$S_1' : \begin{cases} \dot{s} = D(s_{in} - s) - \bar{v}_M(s)\gamma, \\ \dot{p} = \bar{v}_M(s) - \bar{\mu}(p)(p + \bar{c} + 1), \\ \dot{\gamma} = (\bar{\mu}(p) - D)\gamma, \end{cases}$$

where

$$\begin{aligned} \bar{v}_M(s) &:= v_M(s, 1 - \bar{u}), & \bar{v}_R(p) &:= v_R(p, \bar{u}), & \bar{v}_X(p) &:= v_X(p, 1 - \bar{u}), \\ \bar{\mu}(p) &:= \mu(p, \bar{u}), & \bar{c} &:= c(\bar{u}). \end{aligned}$$

We study the local and global stability properties of this reduced system.

Asymptotic behaviour

Local stability

Theorem. Define

$$\bar{f}(p) := \bar{v}_R(p) + \bar{v}_X(p) + \bar{\mu}(p)p = \bar{\mu}(p)(p + \bar{c} + 1),$$

and let p_w be the solution of $\bar{f}(p) = \bar{v}_M(s_{in})$. Then,

- ▶ If $\bar{\mu}(p_w) \geq D$:
 - ▶ The interior equilibrium E_i exists, is unique and locally stable.
 - ▶ The washout equilibrium E_w exists, is unique and locally unstable.
- ▶ If $\bar{\mu}(p_w) < D$:
 - ▶ The interior equilibrium E_i does not exist.
 - ▶ The washout equilibrium E_w exists, is unique and locally stable.
- ▶ The interior equilibrium $E_i := (s_i, p_i, \mathcal{V}_i)$ with

$$\bar{\mu}(p_i) = D, \quad \bar{v}_M(s_i) = \bar{f}(p_i), \quad \mathcal{V}_i := \frac{D(s_{in} - s_i)}{\bar{v}_M(s_i)}.$$

- ▶ The washout equilibrium is $E_w := (s_{in}, p_w, 0)$.

Asymptotic behaviour

Sketch of proof. If $\bar{\mu}(p_w) \geq D$,

$$J_i = \begin{bmatrix} -D - \bar{v}'_M(s)\gamma_i & 0 & -\bar{v}_M(s) \\ \bar{v}'_M(s) & \bar{f}'(p) & 0 \\ 0 & \bar{\mu}'(p)\gamma_i & 0 \end{bmatrix},$$

and the characteristic polynomial is

$$P_i(\lambda) = (\lambda + D)(\lambda^2 + \lambda \underbrace{(D + \bar{v}'_M(s)\gamma_i + (p + \bar{c} + 1)\bar{\mu}'(p))}_{>0} + \underbrace{\bar{v}'_M(s)\gamma_i(p + \bar{c} + 1)\bar{\mu}'(p)}_{>0})$$

which, by Routh-Hurwitz criterion, implies that all eigenvalues have negative real part. If $\bar{\mu}(p_w) < D$,

$$J_w = \begin{bmatrix} -D & 0 & -\bar{c} \\ \bar{v}'_M(s) & -\bar{f}'(p) & 0 \\ 0 & 0 & \bar{\mu}(p_w) - D \end{bmatrix}$$

with characteristic polynomial

$$P_w(\lambda) = (\lambda + D)(\lambda + \bar{f}'(p))(\lambda - \bar{\mu}(p_w) + D). \quad \square$$

Asymptotic behaviour

Global analysis

Proposition. The ω -limit set of any solution of the limiting system S_1' lies in the set

$$\Omega' := \left\{ (s, p, \mathcal{V}) \in \mathbf{R}^3 : s + (p + \bar{c} + 1)\mathcal{V} = s_{in} \right\}$$

As $s = s_{in} - \mathcal{V}(p + \bar{c} + 1)$, one can further reduce S_1' to

$$S_1'' : \begin{cases} \dot{p} = \bar{v}_M(s(\cdot)) - \bar{\mu}(p)(p + \bar{c} + 1), \\ \dot{\mathcal{V}} = (\bar{\mu}(p) - D)\mathcal{V}. \end{cases}$$

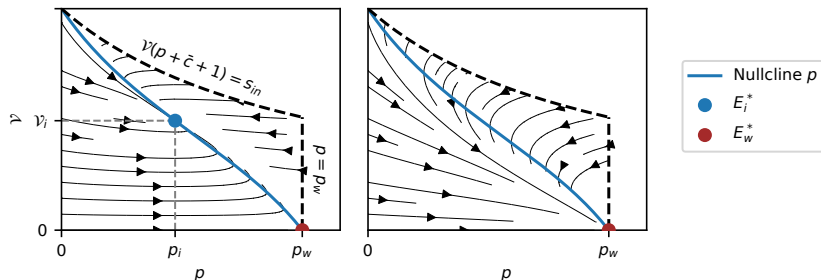
Proposition. Every solution of the limiting system S_1'' converges to

- ▶ $E_i^* := (p_i, \mathcal{V}_i)$ if $\bar{\mu}(p_w) \geq D$,
- ▶ $E_w^* := (p_w, 0)$ if $\bar{\mu}(p_w) < D$.

Sketch of proof. Bendixson-Dulac criterion allows to discard periodic orbits and cycles of equilibria as

$$\begin{aligned} \frac{\partial}{\partial p} \dot{p} + \frac{\partial}{\partial \mathcal{V}} \dot{\mathcal{V}} &= \frac{\partial}{\partial p} \bar{v}_M(s(\cdot)) - \bar{\mu}(p) - \bar{\mu}'(p)(p + \bar{c} + 1) + \bar{\mu}(p) - D \\ &= \bar{v}'_M(s(\cdot)) \frac{\partial s(\cdot)}{\partial p} - \bar{\mu}'(p)(p + \bar{c} + 1) - D < 0. \quad \square \end{aligned}$$

Asymptotic behaviour



Results on asymptotically autonomous systems (Thieme'92) eventually allow to relate the asymptotic behaviour of the 2-dimensional limiting system S_1'' to the behaviour of the full 5-dimensional system S_1 .

Theorem. Every solution of system S_1 converges to

- ▶ the extended interior equilibrium $\hat{E}_i := (s_i, p_i, \bar{u}, x_i, v_i)$, $x_i := \bar{c}v_i$, if $\bar{\mu}(p_w) \geq D$,
- ▶ the extended washout equilibrium $\hat{E}_w := (s_{in}, p_w, \bar{u}, 0, 0)$ otherwise.

Dynamic optimisation

Metabolite production

Maximisation of the metabolite X is considered over a finite time horizon:

$$\text{maximize } D\mathcal{V}_{\text{ext}} \int_0^T x(t) dt,$$

resorting to $(q := (s, p, r, x, \mathcal{V}) \in \mathbf{R}^5)$

$$(\text{OCP}) : \begin{cases} \text{maximize} & \int_0^T x(t) dt \\ \text{subject to} & \dot{q}(t) = F(q(t), u(t)), \\ & q(0) = q_0, \quad q(T) \text{ free}, \\ & u(t) \in [0, 1]. \end{cases}$$

Proposition. Existence holds.

Dynamic optimisation

Extremal flow

We write the (normal) Hamiltonian in terms of state $q := (s, p, r, x, \mathcal{V})$ and costate $\lambda := (\lambda_s, \lambda_p, \lambda_r, \lambda_x, \lambda_{\mathcal{V}})$:

$$H(q, \lambda, u) = -\lambda^0 x + \langle \lambda, F(q, u) \rangle,$$

where F is the right-hand side of S_1 . As the dynamics is control-affine, $H = H_0 + uH_1$ with

$$\begin{aligned} H_0 &= \lambda_s (D(s_{in} - s) - v_M(s, 1 - r)\mathcal{V}) + \lambda_p \left(v_M(s, 1 - r) - v_X(p, 1 - r) - \mu(p, r)(p + 1) \right) \\ &\quad - r\lambda_r \mu(p, r) + \lambda_x (v_X(p, 1 - r)\mathcal{V} - Dx) + \lambda_{\mathcal{V}} (\mu(p, r) - D)\mathcal{V} - \lambda^0 x, \\ H_1 &= \lambda_r \mu(p, r). \end{aligned}$$

By virtue of Pontrjagin's maximum principle, a.e. one has

$$u(t) = \begin{cases} 0 & \text{if } H_1(q(t), \lambda(t)) < 0, \\ 1 & \text{if } H_1(q(t), \lambda(t)) > 0, \end{cases}$$

and the control is singular if H_1 vanishes along a whole subarc.

Dynamic optimisation

Singular arcs

Along a singular arc

$$0 = \dot{H}_1 = \frac{\partial H_1}{\partial q} \dot{q} + \frac{\partial H_1}{\partial \lambda} \dot{\lambda} = \sum_{i=1}^n \left(\frac{\partial H}{\partial \lambda_i} \frac{\partial H_1}{\partial q_i} - \frac{\partial H}{\partial q_i} \frac{\partial H_1}{\partial \lambda_i} \right) = \{H, H_1\} = \{H_0, H_1\} := H_{01}$$

and

$$0 = \dot{H}_{01} = H_{001} + uH_{101}.$$

Proposition. Singular arcs are at least of order two.

$$\begin{aligned} H_{101} = & \mu(p, r) \left[\dot{\lambda}_r \mu_r(p, r) + \lambda_r \frac{\partial}{\partial r} \left(\mu_r(p, r) \dot{r} + \mu_p(p, r) \dot{p} \right) \right] \\ & - \lambda_r \mu_r(p, r) \frac{\partial H_{01}}{\partial \lambda_r} - \lambda_p \mu_p(p, r) \frac{\partial H_{01}}{\partial \lambda_p} \end{aligned}$$

which is also equal to 0 when $H_1 = 0$. $\square$

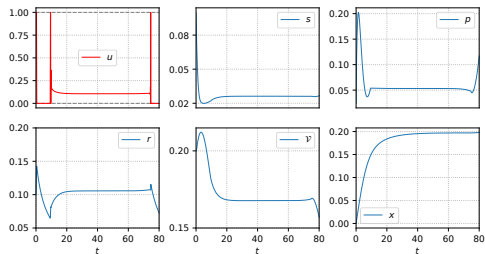
In the case of an order two singular control ($H_{10001} < 0$ along the arc, generalized Legendre-Clebsch condition)

$$u = - \frac{H_{00001}}{H_{10001}}.$$

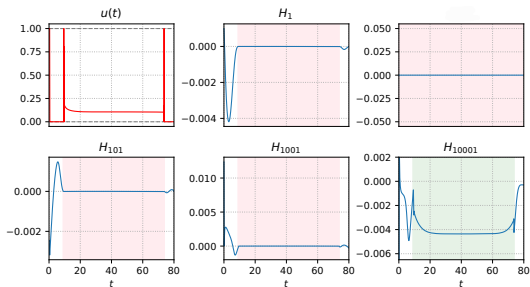
Although the order is *local* (H_{101} is not everywhere zero), one expects connection between bang and singular arcs through Fuller phenomenon.

Dynamic optimisation

Fuller phenomenon



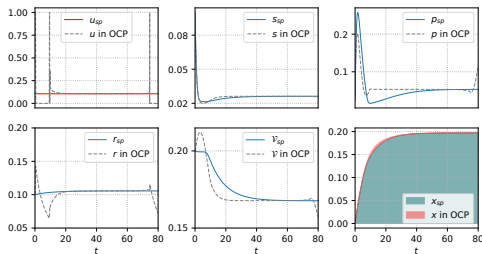
Generalized Legendre-Clebsch condition



Dynamic optimisation

Turnpike property

$$(SP_X) : \begin{cases} \text{maximize} & \bar{x} \\ \text{subject to} & F(\bar{q}, \bar{u}) = 0, \\ & 0 \leq \bar{u} \leq 1. \end{cases}$$



A Lagrange pair $(\bar{q}, \bar{u}, \bar{\lambda})$ of (SP_X) defines a hyperbolic equilibrium of the Hamiltonian singular flow on $\Sigma'' := \{(q, \lambda) \mid H_1 = H_{01} = H_{001} = H_{00001} = 0\}$,

$$H_s(q, \lambda) := H(q, \lambda, u_s(q, \lambda)), \quad u_s(q, \lambda) := -\frac{H_{000001}}{H_{10001}}(q, \lambda).$$

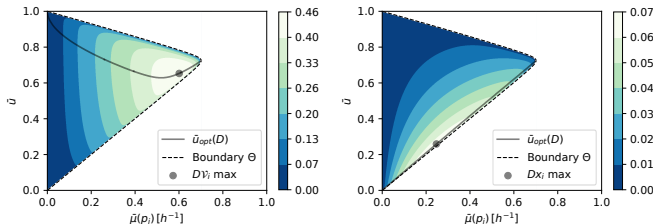
See Trélat-Zuazua'14, Pighin'16 in the regular case, ongoing work with S. Maslovskaya, W. Djema, J.-B. Pomet in the singular case (JOTA 2022).

Static optimisation

Metabolite production vs. biomass maximisation

Maximisation is now performed wrt. the static control $\bar{u}$ plus the dilution rate D of the CSTR:

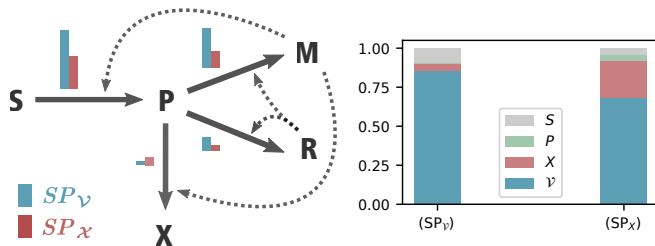
$$(SP_{\mathcal{V}}) : \begin{cases} \text{maximize} & D\bar{\mathcal{V}} \\ \text{subject to} & F(\bar{q}, \bar{u}) = 0, \\ & 0 \leq \bar{u} \leq 1, \end{cases} \quad (SP_X) : \begin{cases} \text{maximize} & D\bar{x} \\ \text{subject to} & F(\bar{q}, \bar{u}) = 0, \\ & 0 \leq \bar{u} \leq 1. \end{cases}$$



Proposition. Any solution of the two static problems lies in the region of existence of the interior equilibrium, $\bar{\mu}(p_w) > D$.

Static optimisation

Discussion



- ▶ 20% decrease of biomass ($\mathcal{V}$) but 500% increase of produced metabolite (X)
- ▶ Difference between static and "gold standard" dynamic optimisation negligible for large fixed final time ($< 1\%$)
- ▶ Richer models: ubiquity of Fuller and (singular) turnpike phenomena
- ▶ Second order conditions in the Fuller case: preliminary approach in Agrachev-Beschastnyi'2020
- ▶ Reproducible research: ct (= control toolbox) initiative control-toolbox.org



The control-toolbox ecosystem gathers Julia packages for mathematical control and applications. It is an outcome of a research initiative supported by the [Centre Inria of Université Côte d'Azur](#) and a sequel to previous developments, notably [Bocop](#) and [Hampath](#). See also: [ct gallery](#). The root package is [OptimalControl.jl](#) which aims to provide tools to solve optimal control problems by direct and indirect methods.

Installation

See the [installation page](#).

Getting started

To solve your first optimal control problem using `OptimalControl.jl` package, please visit our [basic example tutorial](#) or just copy-paste the following piece of code!

```
using OptimalControl

@def ocp begin
    t ∈ [ 0, 1 ], time
    x ∈ ℝ², state
    u ∈ ℝ, control
    x(0) == [ -1, 0 ]
    x(1) == [ 0, 0 ]
    ẋ(t) == [ x₂(t), u(t) ]
    J( 0.5u(t)² ) → min
end

sol = solve(ocp)
plot(sol)
```

Free Azat Miftakhov



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